AI that computes with time and is already winning

A new family of learning machines, built for a world that happens as events, beats the best published models on public benchmarks with a fraction of their size.

5 of 5
datasets won on the public EasyTPP event-prediction leaderboard

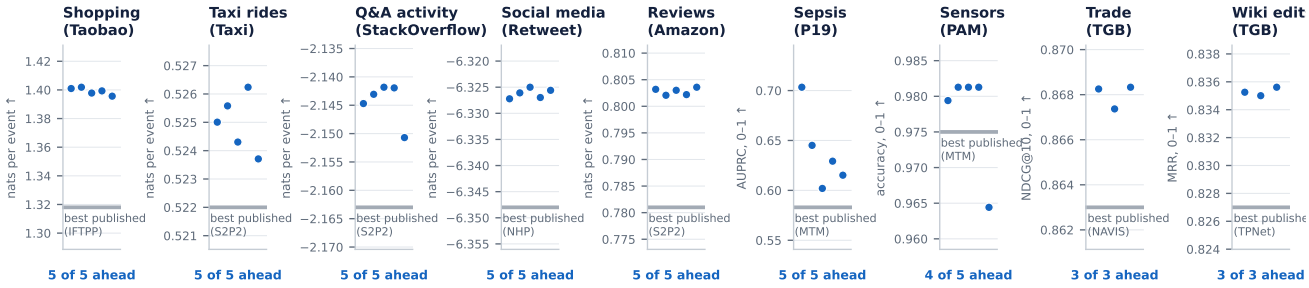
0.639
sepsis early-warning AUPRC vs 0.583 for the best published model ↑

15×
smaller and cheaper than the state-of-the-art model on Retweet, while more accurate

40 / 41
independent runs ahead of the best published result on the nine benchmarks won

In one sentence. Sleeping Machines is building a general-purpose computing substrate for machine intelligence, one that learns through time and sparse events, and its first public results are wins against the strongest published models, achieved by small, efficient models under strict, pre-registered test protocols.

40 of 41 independent runs ahead of the best published result



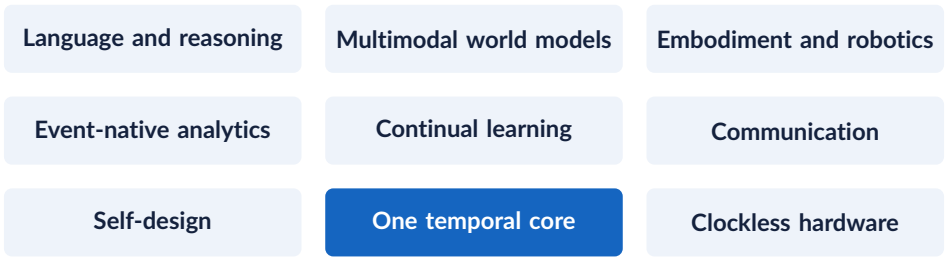
The benchmark wins use completed runs on official data, scored once on sealed test sets. The forward-credit development page reports separate private TRAIN-tail experiments, with its scope stated beside the numbers. Each benchmark's unit, and whether higher or lower is better, is stated beside every comparison (↑ higher is better, ↓ lower is better). Sources: `experiments/B1_EASYTPP.md`, `experiments/B2_IRREGULAR_TS.md`, `report/I_SCIENCE.md`.

THE AMBITION

One substrate for machine intelligence

Today's AI runs on a clock: every input is chopped into fixed steps, and the whole model works at every step whether or not anything happened. The real world is not like that. Purchases, heartbeats, sensor readings, messages, machine logs and words arrive as *events*, irregularly, with meaning in their timing and in their silences.

Sleeping Machines builds learning machines in which **time itself does part of the computation**. Our goal is one trainable substrate that spans:



What makes it different, in plain terms:

- **Memories that live through silence.** Each memory keeps evolving between events, so "how long since" and "what happened in between" are information, not missing data.
- **A race of clocks decides what happens next.** Possible next events race against each other; the one that fires first wins. This gives exact predictions of both *when* and *what*, and the same mathematics reproduces the attention mechanism inside today's large language models.
- **Only the relevant parts wake up.** Updates are sparse and addressed: the machine can hold far more capacity than it uses for any single event, which is where the name comes from.
- **Learning from roads not taken.** The model is also credited for the alternatives that lost the race, so it learns which routes to take.

> **Why this matters commercially.** Most of the world's valuable data is timestamped events: transactions, clinical records, telemetry, security and IT logs, user behaviour. Models that understand time natively can be more accurate on that data and much cheaper to run, on CPUs and edge devices as well as in datacentres. The same principles point to hardware without a global clock, where idle capacity costs almost nothing.

THE PROVING GROUND

The benchmarks, in plain language

We compete on established public benchmarks, using their official data splits and scoring rules, against the best results other teams have published. The headline event benchmark is **EasyTPP** (ICLR 2024), the standard test suite for predicting *when* the next event happens and *what* it will be. Its current leader is **S2P2** (NeurIPS 2025, GE HealthCare and UC Irvine), a modern state-space model. The **Temporal Graph Benchmark** (TGB, NeurIPS 2023) tests who interacts with whom, and when, against temporal graph neural networks.

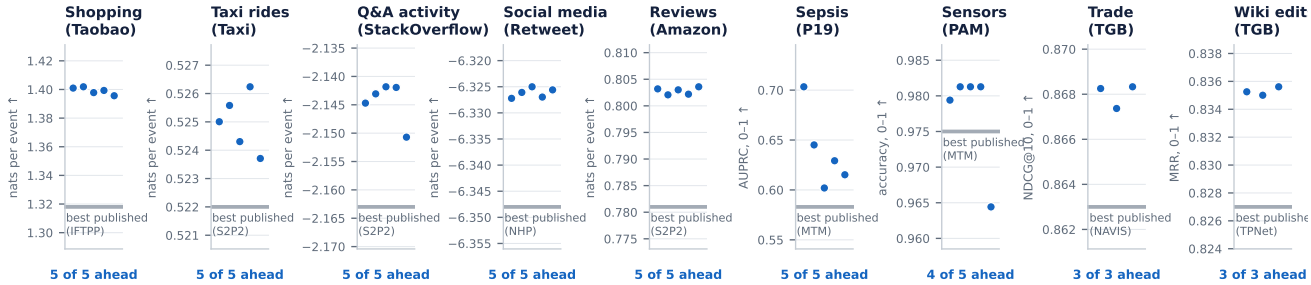
Benchmark	What the data is	Why it matters	Score (unit, direction)
Taobao	Clicks and purchases of users on Alibaba's Taobao marketplace, by product category	E-commerce: anticipating what a shopper does next and when	log-likelihood, nats per event ↑
Taxi	New York taxi pick-ups and drop-offs across city districts	Mobility and logistics: demand and fleet planning	log-likelihood, nats per event ↑
StackOverflow	Badges earned by users of the StackOverflow Q&A site	User engagement and lifecycle prediction	log-likelihood, nats per event ↑
Retweet	Retweet cascades on Twitter, by type of user retweeting	Social media: how information spreads, and how fast	log-likelihood, nats per event ↑
Amazon	Product reviews by Amazon customers, by category	Retail behaviour across product lines	log-likelihood, nats per event ↑
P19 sepsis	PhysioNet 2019: 38,803 intensive-care stays, 34 vital signs and lab values measured at irregular times	Early warning of sepsis, a leading cause of hospital deaths; every hour of earlier treatment matters	AUPRC and AUROC, 0 to 1 ↑
PAM	PAMAP2 physical-activity monitoring: 5,333 recordings from wearable inertial sensors (17 channels), 8 activities	Wearables and health monitoring: recognising what a person is doing from body-worn sensors	accuracy and F1, 0 to 1 ↑
P12 mortality	PhysioNet 2012: 11,988 intensive-care stays, 36 irregular measurements	Predicting in-hospital mortality from the first 48 hours	AUROC and AUPRC, 0 to 1 ↑
TGB trade	UN trade flows between 255 countries, 1986–2016	Which partners a country trades with most next year	NDCG@10, 0 to 1 ↑
TGB wiki	157,474 timed Wikipedia edits by 8,227 users on 1,000 pages	Which page a user edits next	MRR, 0 to 1 ↑
Neural-TPP benchmark (TMLR 2023/2025)	Seven event datasets (LastFM listening, MOOC course activity, Github, Stack Overflow, Wikipedia edits, MIMIC2 hospital visits, Retweets), five fixed splits each	A second, independent event benchmark: when the next event happens and what it is	negative log-likelihood, nats per sequence ↓

Reading the scores.

- **Log-likelihood (nats per event), higher is better.** It measures how much probability the model gave to what actually happened, both its timing and its type. A gain of 0.08 nats per event means the model assigns about 8% more probability to every real event, on average, compounding over every event in a sequence. Values can be negative; less negative is better.
- **AUPRC, higher is better (0 to 1).** How well the model ranks the rare patients who develop sepsis (4% of stays) ahead of everyone else, with emphasis on raising true alarms without false ones. It is the clinically relevant metric when the condition is rare. **AUROC** (0 to 1, higher is better) measures overall ranking.
- **Negative log-likelihood per sequence, lower is better.** The same probability measure summed over a whole sequence and negated, as the neural-TPP benchmark reports it; time and type ("mark") parts are reported separately and add up to the total.
- **NDCG@10 and MRR, higher is better (0 to 1).** How well the model ranks the true next partners or pages near the top of a list of all candidates.
- **Size and cost, lower is better.** Parameters measure how big a model is; multiply-adds per event measure the work needed for each prediction, which drives running cost and energy.

THE WINS

Ahead of the best published models



Each blue dot is one independent run (EasyTPP: five training seeds; sepsis and wearables: the five official data splits; temporal graphs: three pre-registered seeds), scored once on the sealed test set. The grey line is the best published result. Higher is better in every panel: log-likelihood in nats per event for the EasyTPP datasets, AUPRC (0 to 1) for sepsis, accuracy (0 to 1) for wearables, NDCG@10 and MRR (0 to 1) for the two temporal-graph benchmarks.

Benchmark	Best published	Sleeping Machines	Margin	Our size and compute vs the leading model
Taobao nats/event ↑	1.318 (IFTPP)	**1.399 ± 0.003**	+0.081	0.92× S2P2's compute per event; best time and type predictions
Taxi nats/event ↑	0.522 (S2P2)	**0.525 ± 0.001**	+0.003	**1/12 of S2P2's size and compute**; reproduced independently on separate hardware
StackOverflow nats/event ↑	-2.163 (S2P2)	-2.144 ± 0.004	+0.019	1.26× compute; a model of S2P2's size also wins (-2.153)
Retweet nats/event ↑	-6.348 (NHP)	-6.326 ± 0.001	+0.022	**1/15 of S2P2's size and compute**; S2P2 scores -6.365
Amazon nats/event ↑	0.781 (S2P2)	**0.803 ± 0.001**	+0.022	**0.29× S2P2's compute**, 3.6× smaller; best time and type predictions
Wearables, PAM accuracy ↑	0.975 (MTM)	**0.978 ± 0.007**	+0.003	**19× smaller** than MTM (46,316 vs 873K parameters); F1 0.980 vs 0.976; 4 of 5 splits ahead
Sepsis, P19 AUPRC ↑	0.583 (MTM)	**0.639 ± 0.039**	+0.056	62,681 parameters; AUROC 0.916 vs 0.903 ↑; without its temporal memory the same network drops to 0.572 on every split
Trade graphs, TGB tgbn-trade NDCG@10 ↑	0.863 (NAVIS)	**0.868 ± 0.0005**	+0.005	2,107 parameters (NAVIS 1,280); 3 of 3 sealed seeds ahead; a new domain
Wikipedia edits, TGB tgbl-wiki MRR ↑	0.827 (TPNet)	**0.835 ± 0.0003**	+0.008	7,995 parameters; 3 of 3 sealed seeds ahead; 4.5 ms per query on one CPU thread

± is the standard deviation across the runs (five; three on TGB). Compute is multiply-adds per event, counted from S2P2's released code at its published configuration.

One model for the whole benchmark. A single configuration of our model, with no per-dataset tuning, is ahead of the best published result on all five EasyTPP datasets: Taobao 1.397 vs 1.318, Taxi 0.526 vs 0.522, StackOverflow -2.144 vs -2.163, Retweet -6.324 vs -6.348, Amazon 0.802 vs 0.781 (nats per event ↑; 24 of 25 runs ahead; pre-registered protocol), at 0.07× to 1.03× the leading model's compute. Across all nine benchmarks, 40 of 41 independent runs are ahead.

BEYOND EASYTPP

The same model wins on two more datasets

One model configuration won all five EasyTPP datasets. Applied unchanged to a second event benchmark, it also beats the best published single-model total on **Github activity and LastFM listening**. These results use the independent neural temporal-point-process benchmark of Bosser & Ben Taieb (TMLR 2023, 2025): the same configuration, trained on each dataset, with no dataset-specific tuning.

Total prediction loss nats/sequence ↓	Best published model	Sleeping Machines	Improvement
Github activity	-269.70 (LNM++)	-270.62	**0.92**
LastFM listening	-697.50 (LNM++)	-774.64	**77.14**

Five official splits; mean total negative log-likelihood per sequence, lower is better. Standard errors across the five splits: Github 48.69; LastFM 48.38.

Why this matters. A model selected on one event benchmark transfers to a second without a dataset-specific search. Github tests software-development activity; LastFM tests listening behaviour. On LastFM the improvement is 77.14 nats per sequence over the strongest published model's total score.

Stronger timing predictions and addressed memory

The component scores show where the temporal architecture contributes:

Component nats/sequence ↓	Best published	Sleeping Machines	Achievement
LastFM: event timing	-1363.78	-1380.05	16.27 lower timing loss
Github: event timing	-382.40	-390.93	8.53 lower timing loss
Stack Overflow: event timing	-91.10	-91.60	0.50 lower timing loss
Wikipedia: event type	144.79	**27.65**	80.9% lower type loss

Memory that copies what it has seen. About 90% of consecutive Wikipedia edits return to the page just edited, while 10–27% of test pages never appear in training. Our addressed memory stores pages as they occur and can reuse them later in the sequence. Its type-prediction loss is **27.65 versus 144.79** for the strongest published type predictor. This is a component achievement; Wikipedia's total failed because two sequences received exploding timing hazards.

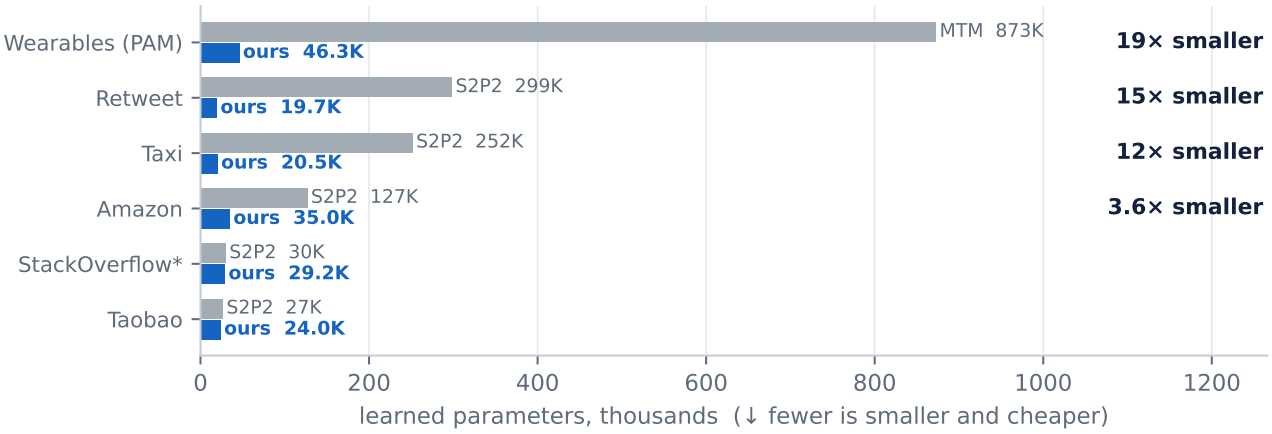
Published models and a separate stretch target

The pre-registered research target combines the best timing score and best type score from *different* published models. No single published model reaches that composite. Github and LastFM beat the strongest published single-model totals while missing this stricter composite target (-272.90 and -849.65, respectively). The original composite-target verdicts are unchanged. The other five datasets do not have total-score wins.

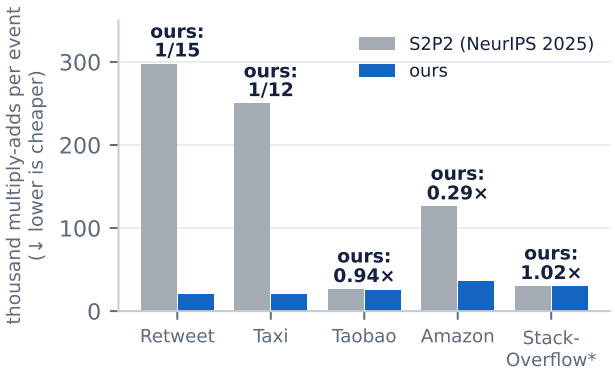
Protocol and complete record. Validation selected checkpoints; five fixed test splits; numerical guarded reruns as specified in B4. Github has 1.3% zero gaps: published log-normal references replace zero by 10^{-20}; our timing model floors at the recording cell. The comparisons above retain those recorded scoring conventions. Full seven-dataset totals, numerical failures, rerun rules and recording-grid audits: `experiments/B4_NTPP_BENCHMARK_PROPOSAL.md` and the report's B4 loss analysis.

EFFICIENCY

Winning while small



Learned parameters, in thousands; fewer means a smaller, cheaper model (lower is better). Grey: S2P2, the leading published state-space model on EasyTPP, at its published configuration for each dataset; for PAM, MTM (2025) at the parameter count it reports for PAM. Every Sleeping Machines model shown is also more accurate than the model beside it. *StackOverflow: the model matched to S2P2's size, which also wins.



Work per prediction, in thousands of multiply-adds per event (lower is cheaper). "ours: 1/15" means our model needs one fifteenth of S2P2's work.

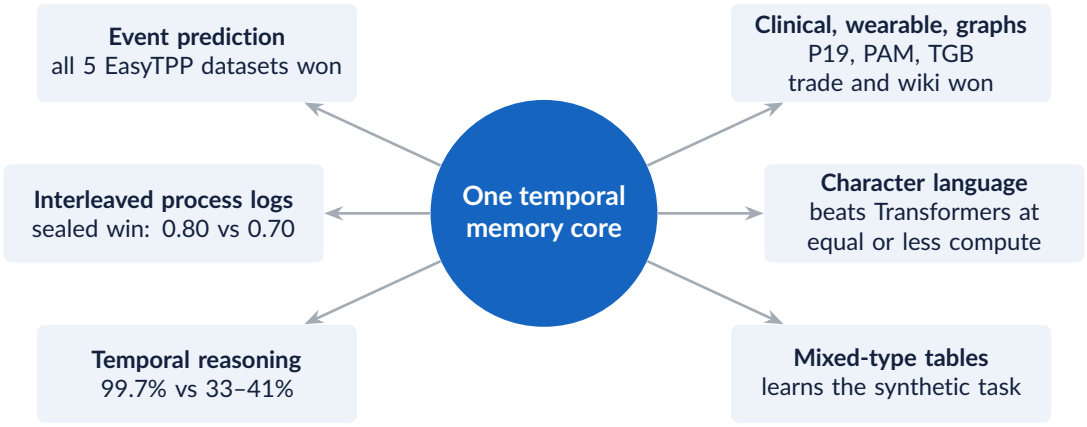
What size buys.

- **Running cost and energy** scale with the work per prediction.
- **Deployment anywhere**: models of 20,000 to 60,000 parameters run on ordinary CPUs, phones and embedded devices, close to the data.
- **Privacy**: clinical and industrial data can stay on site.
- **Headroom**: accuracy at small size means the architecture, not brute scale, is doing the work.

GENERALITY

One family across different kinds of data

S2P2 is shown on one kind of task. Our family wins on four kinds of data (event streams, clinical records, wearable sensors, interaction graphs) with shared machinery: the sepsis classifier reuses *the same temporal memory layer, same code and same size*, as the event-prediction model.



One shared core across four domains (development data, first seed). We trained a *single* temporal memory of 14,080 parameters on taxi rides, online shopping, Q&A activity and product reviews at the same time; only the small input and output layers differ per dataset. It matched or beat the four separately trained models on three of the four (log-likelihood, nats per event, higher is better: Taxi 0.490 vs 0.487, Taobao 1.290 vs 1.284, StackOverflow -2.186 vs -2.177), with about 40% fewer parameters in total. Mobility, shopping and online-community timing share one learned temporal dynamics. Amazon trailed and was still improving when training stopped; per-dataset training budgets are the next test.

Why generality matters for value. A company with one specialised model has one product. A company with a core that transfers across data types can serve many markets from one technology, and every new domain makes the core stronger.

WHY THE LARGER VISION IS ACHIEVABLE

Each layer of evidence supports the next

Next: scale and transfer. Large-scale language on GPUs, skill transfer between data types, clockless hardware.

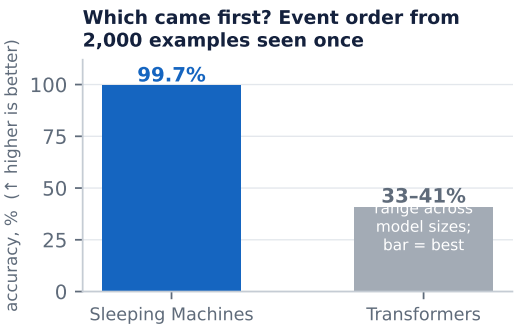
5. It generalises. One shared core across four event domains; one memory layer for events and clinical records.

4. It wins public benchmarks. All 5 EasyTPP datasets, P19 sepsis, PAM wearables and two TGB graph leaderboards, at up to 15× smaller size; 40 of 41 runs ahead.

3. It learns language efficiently. On 10M characters it beats a Transformer: 1.888 vs 1.908 bits per character ↓ with 0.40× its training compute. At 90M characters it beats a tuned Transformer-256×4 (1.783 vs 1.811 ↓) at 1.06× its compute.

2. The mechanisms work. Event order learned from 2,000 examples seen once: 99.7% accuracy vs 33–41% for Transformers ↑. Timing-only discrimination: 95.3% where any order-only model is capped at 50% ↑. Learning from roads not taken improves language 2.507 → 2.371 bits per character ↓ for 0.3% extra training work; more idle capacity improves it further at the same active work.

1. Mathematics: it contains today's leading architectures. A race of clocks reproduces the attention of Transformers *exactly*, and Mamba-style selective state-space models are exact members of the family (verified numerically to 4×10^{-13}). Everything those models can represent, ours can, plus native time and sparsity.



Why the steps add up. The family is not a narrow trick for one benchmark. Its mathematics contains the architectures that power today's AI; its mechanisms solve temporal problems those architectures fail at; it already learns language competitively at small scale; and it wins public benchmarks on four kinds of data with one shared memory design. Scale is the remaining work, and each layer beneath it is already demonstrated.

Hardware. In our cost model, the native language model moves 5.8× fewer bytes per character than a comparable Transformer at better quality (modelled, not yet measured on silicon; a funded milestone).

RIGOR

How we know the results are real

Sealed tests, rules fixed in advance. Every win uses the benchmark's official data splits. The reporting rule is written down before the test set is touched, and each model is scored on the test set exactly once.

The benchmark's own scorer agrees. Re-scoring our models with EasyTPP's own estimation method gives the same results within its sampling noise.

Independent runs. Benchmark headline results are averages of independent runs (five seeds or official splits; three pre-registered seeds on TGB), with its spread: 40 of 41 runs are ahead of the best published result.

We found and closed a scoring pitfall. Timestamps in these datasets are rounded (for example to whole seconds). Flexible models can exploit the rounding for large, fake gains. We built an audit that removes the rounding; we withdrew our own early results that failed it, and every reported win passes it.

Reproduced on separate hardware. The Taxi win was reproduced independently on a separate desktop computer with separately downloaded data: 0.5252 vs 0.5250 nats per event ↑, every run above S2P2. One-command kits let third parties rerun the EasyTPP and both TGB wins.

Losses reported as plainly as wins. Where we are behind, we say so with the numbers and the next test (next page). Comparisons always use the same units and protocol as the published result.

LEARNING MORE ECONOMICALLY

Teaching the model how to learn

The earlier pages show that our models can make strong predictions with small models. We are also tackling the cost of *learning*: deciding which parts of a model should change when new information arrives.

A sparse model takes only a few routes through its available memory and computation. To improve, it needs to learn from those choices, including alternatives it did not take. Trying every alternative can make training expensive even when prediction is cheap.

> **Our approach.** Use what the model discovers while making predictions to help it evaluate alternatives and choose better learning updates. The ambition is a model that learns both *what to predict* and *how to improve*, using the same event-based architecture.

98.4%
less error when estimating the value of alternative learning routes

2.09×
faster evaluation of alternatives in the measured experiment

77.8%
less randomness in the route controller's learning signal

What these results mean.

- **More accurate guidance.** On eight new language samples, information from the model and its learning state estimated the relative value of alternative routes more accurately than a fixed estimate. Every sample improved.
- **Less repeated work.** Reusing this information, with an exact correction, took 12.9 seconds instead of 26.9 seconds for separately evaluating all alternatives. Both measurements include discovering the alternatives.
- **A steadier learning signal.** Random route selection introduces noise into learning. In a separate controller experiment, the corrected signal reduced that source of noise while preserving its average direction.

Significance. These experiments demonstrate that information already produced by a model can make an expensive part of learning more accurate and more economical. They give a concrete engineering basis for our learning architecture. The measured speedup applies to evaluating learning alternatives, not to training an entire model.

Learning rules can themselves improve. In a separate event-prediction experiment, a trained controller adjusted the model's learning updates and improved prediction quality in all nine tested combinations of model depth and initialization. Those fits used more CPU time. This shows that learned update rules can improve quality within our model family.

Status: a working research prototype. The 98.4% result uses eight private language samples; speed uses four samples and three timing repeats; noise uses four samples and a fixed controller. These are separate component measurements. The update-controller result uses three initializations at depths 2/4/8 on Taxi training/development data and retains ordinary error propagation. Detailed methods and completed result hashes are in `investment/forward_credit_evidence_20261011.json`.

ONLINE MARKET PREDICTION WINS

Confirmed on a new month, chosen in advance

Our native model beats both Transformer and LSTM on chronological market forecasts while **all models adapt their weights online**. Each forecast is scored before its outcome is used for learning. The win held on **1 October data that no model had seen**, with every choice fixed and published from August evidence before the October data were acquired.

Lower prediction loss is better. 1 October: same 65,536 consecutive scored trades.

Model	Prediction loss	Online GFLOPs
Sleeping Machines	**8.07854**	**8.51**
LSTM	8.10271	9.51
Transformer (cached)	8.17919	124.68
Transformer (re-encoded)	8.17638	7,963.12

Native wins by 0.02417 nats/event over LSTM and 0.10066 over cached Transformer, and leads in all 16 of 16 consecutive 4,096-trade windows. The scored forecasting-and-learning updates use **14.7 times less arithmetic than the cached Transformer** and 936 times less than the re-encoded one.

Why this matters. Markets change; a forecaster has to keep learning while it predicts. This test carries each model's complete August learner, memory and optimizer state across a five-week calendar gap and lets it adapt on the new month. The ranking and the work advantage both transfer. On the August development stream (65,536 forecasts) the same models score 5.17588 native, 5.19672 LSTM, 5.27966 re-encoded and 5.28043 cached Transformer. Stored weights plus numerical prediction state require **0.146 MB versus 2.334 MB** for the cached Transformer (16 times less, logical tensor bytes at reference precision).

What each model is allowed to do. All receive the same chronological observations and update after 32 revealed outcomes. Native temporal state and LSTM recurrent state persist. The Transformer keeps 2,048 events of history, either as cached past-version keys/values or re-encoded with current weights after every update, plus persistent addressed output memory. No memory reset is imposed at update boundaries or across the calendar gap.

Next. Seeds 1 and 2 are being acquired to turn this single-seed win into a confirmed win; the sealed September test stays unopened until then.

Scope: two-layer, width 32 models; seed 0; Binance BTCUSDT public trades. Acquisition on 24 August TRAIN; August development stream 25 August. October: preregistration committed before data acquisition, 128 context events then 65,536 forecasts, no October tuning. Online arithmetic covers scored-stream forward/backward/optimizer work; original fitting, warmup, context-only preparation and gates are separate; special functions separate. CPU wall time is an implementation diagnostic. No trading-return or measured chip-energy claim. Source-bound results and shared units:
`investment/online_market_evidence.json`.

INDUSTRIAL FAULT DETECTION

A sealed win on interleaved factory logs

A factory log interleaves the events of many items and machines with no item labels. A failing machine shows up as steps that slowly take longer. Our model learns, from normal logs alone, how long each production step takes, links each event to the earlier event it most likely continues, and keeps a separate running anomaly score for every step. **A slowing machine stands out within the first half of a production run.**

Higher is better. Fault detection (AUROC) after 512 events per line, 2,000 normal and 2,000 faulty runs.

Detector	AUROC
Sleeping Machines (three independent fits)	**0.801**
Time-encoded Transformer (three seeds)	0.704
Best of nine classical detectors	0.676
LSTM (three seeds)	0.667

Ahead by 0.097 AUROC (95% interval 0.088 to 0.121); every one of our three fits beats every reference. Wear faults 0.772 versus 0.650; retry faults 0.829 versus 0.758.

Why this matters. Predictive maintenance runs on exactly this kind of data: anonymous, interleaved machine events where timing carries the signal. The model binds events through time, the family's core mechanism: delayed clocks racing to explain each event.

How the test was run. The test data were generated by our open simulator only after the full plan (model, settings, references, decision rule) was committed. Each model was scored once. No item identities are used anywhere; the model's knowledge is fixed during scoring, so it cannot learn a fault as normal. An identity-aware oracle, not eligible, reaches 0.821 on the earlier test.

Scope: FAS v2 simulator (Keski-Valkama 2017), two interleaved assembly lines of 30 items, 2% dropped events; references are the pre-selected Transformer and LSTM configurations of the earlier sealed test plus nine information-matched classical detectors. Source-bound result: `investment/fas_evidence.json`.

LANGUAGE

Addressed memory leads our language models

Our strongest language models give the network explicit, addressed access to its own past: events write keyed messages, and later tokens read the relevant ones directly. This is the same principle behind the fault-detection win on the previous page.

Lower is better. Prediction loss, nats per token, on the same 8,192 FineWeb development tokens (GPT-2 tokens, 1,024-token history).

Model	Prediction loss
Keyed temporal-memory model (two seeds)	**5.60 / 5.62**
Earlier flagship + addressed episodic memory	5.634
Earlier flagship (without addressed memory)	5.978

Addressed memory is worth 0.35 nats per token on the flagship itself; removing the keyed read from the keyed model costs 0.10 at 1M training tokens.

Learning more efficiently. Sparse counterfactual teaching improves prediction by **0.093 nats per token** while cutting complete CPU training time by **15.3%** (6.528 vs 6.621, 7,734 vs 9,129 seconds at 8.39 million tokens). Calibrated local learning controls, where each unit limits its own update without a global step, add a further **0.005 nats** at **+0.41%** CPU time. Local control removes one chip-wide synchronization requirement: the basis for clockless hardware, where independent units run at their own pace (ETH Zürich; DTU). No chip speed or energy gain is claimed yet.

Continuation. Scale the strongest keyed member on fresh tokens. Compare shallow and deep members with addressed reads in every layer; fund larger depth only when it improves prediction over the strongest shallow member. Unchanged race stacking worsened loss from 6.160 (D2) to 6.290 (D8) to 6.372 (D16). The keyed member uses full backpropagation; hard route races and counterfactual route credit remain a separate integration step.

Scope: keyed model 3.33M parameters, repeated epochs over 4M training tokens; flagship and episodic arm single seed; quality placement, not matched compute. Source-bound results: `investment/language_memory_evidence.json`, `investment/language_learning_evidence.json`, `investment/locality_scaling_evidence.json`.

WHAT THE EVIDENCE ADDS UP TO

A proven prediction engine and a developing learning engine

Sleeping Machines aims to be a general-purpose substrate for language and reasoning, multimodal world models, embodiment, continual learning, communication, self-design and hardware. Its evidence already spans several parts of that ambition.

Strong prediction with small models. The public benchmark results lead this brief because they are the strongest evidence: nine benchmark wins across four kinds of data, including all five datasets of the standard event benchmark, with models up to 15 times smaller. Two preregistered confirmations add a new month of market forecasting while every model learns, and a sealed industrial fault-detection test: 0.801 versus 0.704 AUROC for the best Transformer. That gives the model family a measured advantage on specific tasks, using existing computers.

A practical route to improving learning. We have measured more accurate and faster evaluation of learning alternatives, better predictions from learned update rules, and online market learning that stays ahead of LSTM and Transformer learners on unseen data. Together they show working pieces of a machine that learns how to learn. Each result has its own scope; the pieces are not one combined speedup.

A mathematical foundation. We have derived and numerically checked how route-level learning errors depend on prediction uncertainty and differences between alternatives. The direct route formulation can avoid multiplying local estimation errors through successive layers. This explains why the design is worth pursuing; it is an analytical result, separate from measured scaling.

Where we are behind. On P12 mortality the family scores 0.871 versus 0.880 AUROC. On the second event benchmark (page 5), our current model with addressed episodic memory is in development; its verdicts follow when the test splits are scored.

Where scaling stands. The wins above come from small models on event data. Large-scale language is not yet won: our strongest token model scores 5.60 nats per token after 4 million training tokens, and a new addressed episodic memory improves the earlier flagship by 0.35 nats, but added depth does not yet pay for itself. A two-seed depth ladder is running now; the next decisive steps are deeper models that learn their new layers and faster execution than the current single-thread CPU training.

The commercial significance. The public wins support the value of small, accurate models for timestamped data in industry, IT operations, security and finance; the sealed fault-detection win and the new-month market win show it on predictive maintenance and live market data. The learning work addresses a further source of cost: keeping models useful as new information arrives. If the measured component advantages carry through to complete applications, they could make continual adaptation more affordable on local machines and support sparse, clockless learning hardware. That broader opportunity is the significance of the design, rather than a measured revenue, energy or whole-model training claim.